EurJOC
European Journal of Organic Chemistry

Full Paper
doi.org/10.1002/ejoc.202000851

Chemistry Europe
European Chemical Societies Publishing

Gold Catalysis

Gold-Catalyzed Sequential Cycloisomerization of 1,3,9-Ene-allene-ynes to Fused Spirocarbocycles

Xianxiao Chen,[a][‡] Xiangxiang Ling,[a][‡] Di An,[a] and Weidong Rao*[a]

Abstract: An efficient synthetic method to prepare architecturally challenging spiro tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalene derivatives from Au(I)-catalyzed intramolecular sequential cycloisomerization of linear 1,3,9-ene-allene-ynes is described. The proposed cycloisomerization pathway involves an initial gold-catalyzed 1,3-ene-allene cycloisomerization, followed by a formal [4+2] cycloaddition to give fused spirocarbocyclic products containing a quaternary centre.

Introduction

Fused spriocarbocycles are common structural motifs found in a diverse range of bioactive terpenoid natural products and pharmaceuticals.[1–7] For instance, conidigenone,[2] conidigenone C,[3] waihoensene,[4] phomopsene,[5] lycojaponicumin C[6] and roquefornine A,[7] all feature 6–5–5 fused spiro-tricyclic carbocycle skeletons (Figure 1). Moreover, they also serve as versatile synthetic precursors for the synthesis of structurally complex natural products.[8] Accordingly, the development of new and efficient synthetic strategies for their efficient construction from readily accessible substrates would be highly desirable.

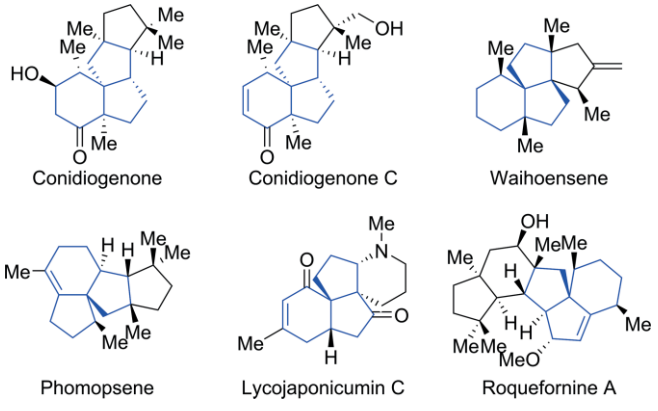


Figure 1. Selected examples of natural products containing the 6–5–5 fused spiro-tricyclic carbocycle skeletons.

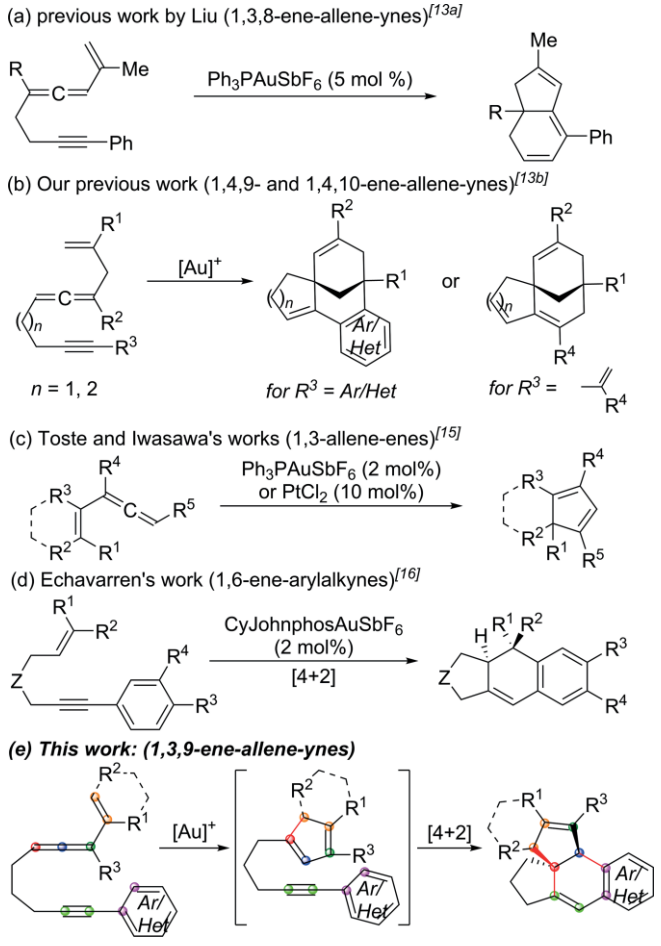
[a] *X. Chen, X. Ling, D. An, Prof. Dr. Weidong Rao*
Jiangsu Provincial Key Lab for the Chemistry and Utilization of Agro-Forest Biomass, Jiangsu Co-Innovation Center for Efficient Processing and Utilization of Forest Resources, College and Chemical Engineering, Nanjing Forestry University, Nanjing 210037, China
E-mail: weidong@njfu.edu.cn
http://hg.njfu.edu.cn/info.php?cid=25
[‡] These authors contributed equally to this work.
Supporting information and ORCID(s) from the author(s) for this article are available on the WWW under https://doi.org/10.1002/ejoc.202000851.

Recently, gold-catalyzed cycloisomerization of enynes, ene-allenes and allene-alkynes have emerged as one of the most powerful and versatile synthetic strategies for the efficient and atom-economical assembly of various carbo- and heterocycles from readily accessible substrates in a single operation.[9–11] However, the analogous cascade reactions of starting materials containing an alkene, allene and alkyne motif, such as 1,n,m-ene-allene-ynes, have been rarely explored.[12,13] The exploration into these reactions has been limited, so far, their in situ generation as reactive intermediates from Au(I)-catalyzed tandem 1,3-acyloxy migration/cycloisomerization of 1-ene-n,m-diyne esters.[14] To the best of our knowledge, only two reports have so far been developed on the exploration of the reactivities of preformed 1,n,m-ene-allene-ynes in gold catalysis, as shown in Scheme 1.[13] For example, Liu and co-workers disclosed that 1,3,8-ene-allene-ynes could proceed efficiently in the presence of gold catalyst to form bicyclo[4.3.0]nonadiene derivatives through 6-endo-dig/Nazarov cyclization cascade (Scheme 1a).[13a] Very recently, we developed an efficient synthetic protocol to prepare cycloalkyl- and (hetero)aryl-fused bicyclo[3.3.1]nonane derivatives that relied on gold(I)-catalyzed cyclopropanation/[5+3] cycloaddition of 1,4,9- and 1,4,10-allenenynes (Scheme 1b).[13b]

In 2007, two novel and efficient synthetic approaches that relied on gold(I) and $PtCl_2$-catalyzed cycloisomerizations of 1,3-ene-allenes leading to highly functionalized cyclopentadienes were reported independently by Toste[15a] and Iwasawa[15b] (Scheme 1c). In addition, Echavarren and co-workers developed a novel gold(I)-catalyzed intramolecular [4+2] cycloaddition of 1,6-arylalkyne-alkenes which provided rapid access to 2,3,9,9a-tetrahydro-*1H*-cyclopenta[b]naphthalene derivatives (Scheme 1d).[16] Inspired by these pioneer works and as part of ongoing efforts in the field of gold catalysis,[13b,14] we became interested in the potential reactivity of preformed 1,3,9-ene-allene-ynes, the chemistry of which has so far remained unexplored. We envisioned that a gold(I)-cascade process involving an initial 1,3-ene-allene cycloisomerization followed by a formal [4+2] cycloaddition[17] might be feasible using 1,3,9-ene-allene-ynes. Herein, we describe the details of this chemistry that pro-

© 2020 Wiley-VCH GmbH

Full Paper
doi.org/10.1002/ejoc.202000851



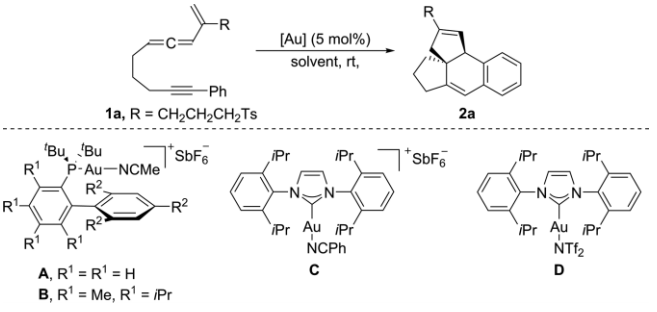
Scheme 1. Gold-catalyzed intramolecular cycloisomerization of preformed 1,n,m-ene-allene-ynes and pioneer work involving 1,3-allene-enes and 1,6-ene-arylalkynes.

vides an expedient and chemoselective approach to access an architecturally challenging class of spiro tetrahydro-*3H*-dicyclopenta[*a,b*]naphthalene frameworks containing an all carbon quaternary centre from all-carbon tethered linear 1,3,9-ene-allene-ynes.

Results and Discussion

Our studies commenced by examining the gold(I)-catalyzed sequential cycloisomerization of 1,3,9-ene-allene-yne **1a** to establish the optimum reaction conditions (Table 1). This initially revealed that treating **1a** with 5 mol-% of gold(I) phosphine catalyst **A** in DCE at room temperature for 12 h afforded the sprio tetrahydro-*3H*-dicyclopenta[*a,b*]naphthalene **2a** in 86 % yield (entry 1). The structure of **2a** was unambiguously ascertained by X-ray crystallographic analysis of a closely related analogue (vide infra).[18] A similar outcome was observed upon changing catalyst from **A** to more sterically crowded Au(I) complex **B** or *NHC*-gold(I) (*NHC* = N-heterocyclic carbene) complexes **C-D** with **C** providing the highest yield of 90 % (entries 2–4). However, no reaction was detected by TLC and ^{1}H NMR measurements when $Ph_3PAuNTf_2$ was used (entry 5). Switching the reaction solvent from DCE to PhMe resulted in a slightly lower yield of **2a** (entry 6). However, no reaction was observed for reactions using THF or MeCN as solvents (entries 7–8). Therefore, reaction of **1a** in the presence of gold catalyst **C** (5 mol-%) in DCE at room temperature for 12 h provided the optimum reaction conditions.

Table 1. Optimization of Reaction Conditions.[a]



Entry	Catalyst	Solvent	Yield [%][b]
1	**A**	DCE	86
2	**B**	DCE	88
3	**C**	**DCE**	**90**
4	**D**	DCE	87
5	**$Ph_3PAuNTf_2$**	DCE	–
6	**C**	PhMe	82
7[c]	**C**	THF	–
8[c]	**C**	CH_3CN	–

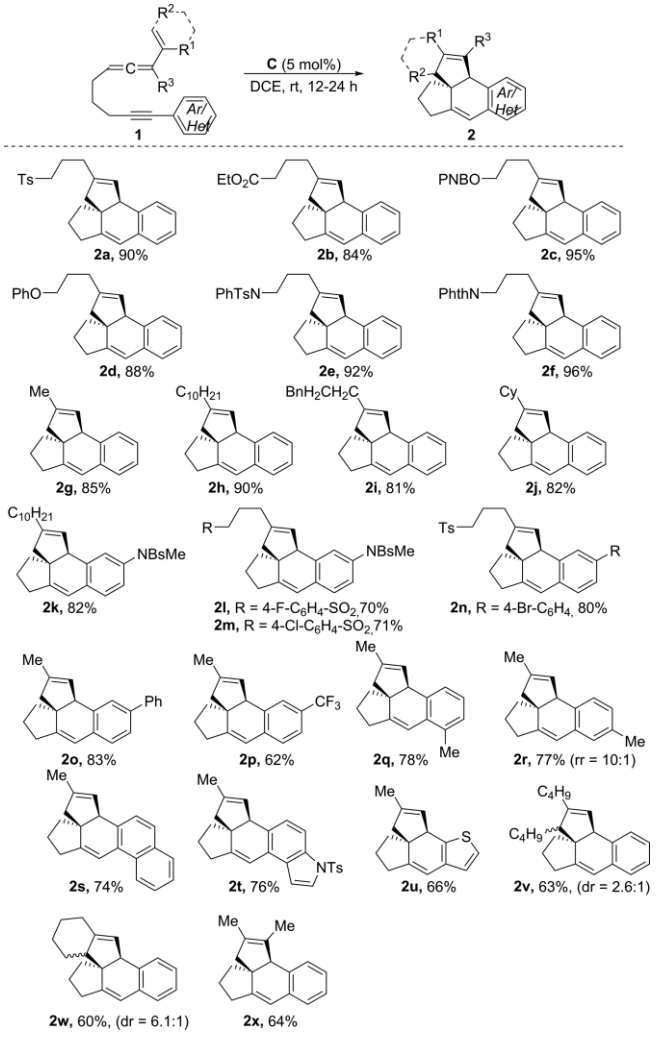
[a] All reactions were performed using **1a** (0.3 mmol) and gold catalyst (5 mol-%) in solvent (3 mL) at room temperature for 12 h. [b] Isolated yield. [c] No reaction based on TLC analysis and ^{1}H NMR measurements of the crude reaction mixture.

With the optimized conditions in hand, we next examined the scope and generality of the present procedure with a variety of 1,3,9-ene-allene-ynes, and the results are outlined in Scheme 2. Overall, the *NHC*-gold(I) complex **C**-catalyzed reaction conditions proved to be broad, producing a diverse array of sprio tetrahydro-*3H*-dicyclopenta[*a,b*]naphthalenes containing a variety of substitution patterns in 60–96 % yields. Reactions of substrates containing a variety of functional groups, such as esters (**1b**–**c**), ether (**1d**), sulfonamide (**1e**), phthalimide (**1f**) on the alkenyl side chain were well-tolerated, furnishing the corresponding products **2b**–**f** in excellent yields. Similarly, substrates possessing either linear (**1g**–**i**) or cyclic alkyl groups such as cyclohexyl (**1j**) at the R^1 position all proceed smoothly under the optimal conditions and led to the formation of the desired products **2g–j** in 81–90 % yields. Subsequently, substrates with a pendant aryl acetylenic motif containing different substituents, such as sulfonamide (**1k**–**m**), aryl (**1n–o**) at the *para*-position or methyl (**1q**) at the *ortho*-position of the benzene ring were found to be amenable, providing the desired products **2k–o** and **2q** in good yields (70–83 %) with the structure of **2o** unambiguously confirmed by X-ray crystallographic analysis.[18] However, a slightly lower yield of 62 % was observed when the substrate **1p** containing a strong electron-withdrawing CF_3 group at the *para*-position of the phenyl moiety was subjected to the optimized conditions. Notably, substrate **1r** with a *meta*-methyl phenyl at the alkyne terminus provided the desired product **2r** in 77 % yield as an inseparable

© 2020 Wiley-VCH GmbH

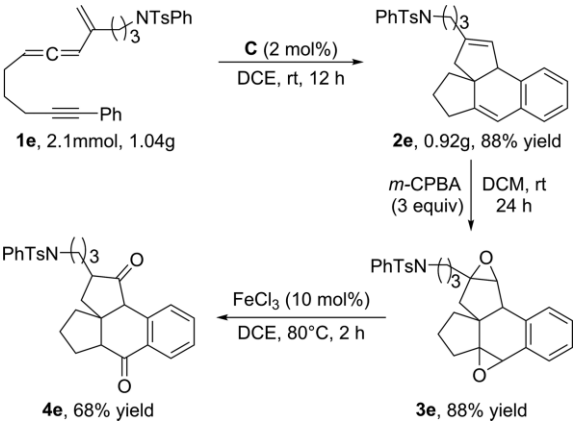
Full Paper
doi.org/10.1002/ejoc.202000851

mixture of regioisomers in a ratio of 10:1. Intriguingly, we found that the reactions of 1,3,9-ene-allene-ynes bearing a pendant heteroaryl moiety at the alkyne terminus, such as indole (**1t**) and thiophene (**1u**), proceeded efficiently to produce the corresponding products **2t–u** in 66–76 % yields. Significantly, sequential cycloisomerization of substrates incorporating disubstituted alkenes (**1v–w**) was also compatible, affording tetrahydro-3*H*-dicyclopenta[*a,b*] naphthalenes **2v–w** in good yields, albeit as inseparable mixtures of diastereomers in 2.6:1 and 6.1:1 ratios for **2v** and **2w**, respectively. Gratifyingly, the starting material that bears a trisubstituted allenyl motif (**1x**) also proceeded well to provide the corresponding product **2x** in 64 % yield.



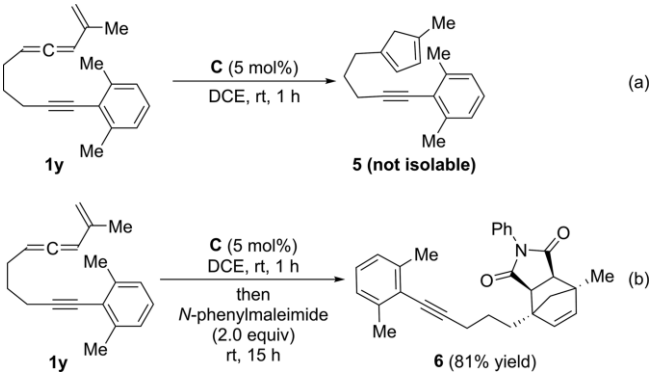
Scheme 2. Substrate scope of 1,3,9-ene-allene-ynes. Reaction conditions: **1** (0.3 mol) and gold complex **C** (5 mol-%) in DCE (3 mL) at room temperature for 12–24 h; isolated yields.

To further demonstrate the scalability and practicality of the present protocol, a scale-up reaction of **1e** (2.1 mmol, 1.04 g) was carried out using only 2 mol-% gold(I) catalyst **C** under the optimized conditions, affording the desired product **2e** in excellent yield of 88 % (Scheme 3). Subjecting compound **2e** to *m*-CPBA in DCE at room temperature for 24 h provided the diepoxide product **3e** in 88 % yield as a single diastereomer. Subsequently, treatment of **3e** with $FeCl_3$ in DCE at 80 °C for 2 h furnished the diketone **4e** in 68 % yield.



Scheme 3. Gram-scale synthesis and synthetic transformations.

Several control experiments were performed to gain further insight into the reaction mechanism (Scheme 4). First, to isolate a possible intermediate, substrate **1y** that bears a 2,6-dimethylphenyl motif at the alkyne terminus was subjected to the optimized reaction conditions (Scheme 4a). Unfortunately, the cyclopentadiene product was not isolable due to rapid polymerization upon treating with column chromatography. Gratifyingly, repeating the reaction for 1 h followed by the addition of *N*-phenylmaleimide for an additional 15 hours produced the [4+2] adduct **6** in 81 % yield (Scheme 4b).[19] These results indicated that the cyclopentadiene species was involved as the key intermediate.



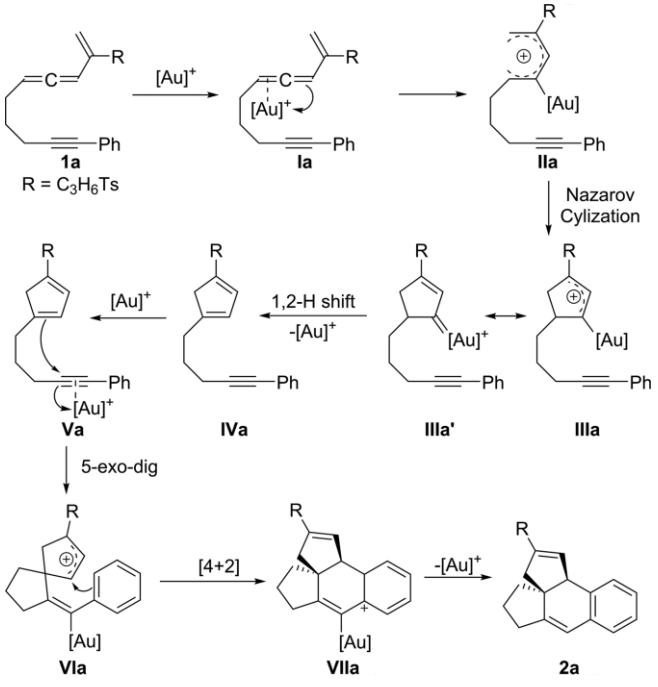
Scheme 4. Control experiments.

On the basis of the above observations and literature reports,[15,16] a plausible mechanism for the present Au(I)-catalyzed sequential cycloisomerization reaction is proposed, as illustrated in Scheme 5. Using **1a** as a representative example, this might initially involve selective activation of the allene moiety of the substrate by the gold(I) complex to give the gold-coordinated 1,3,9-ene-allene-yne species **Ia.** This may then lead to a Nazarov cyclization of **Ia** via the pentadienyl cation **IIa** to give the cationic intermediate species **IIIa** or its resonance form

© 2020 Wiley-VCH GmbH

Full Paper
doi.org/10.1002/ejoc.202000851

gold carbenoid species **IIIa'**, followed by 1,2-hydride migration and deauration to produce the cyclopentadiene intermediate **IVa** and regenerate the gold(I) catalyst.[11a,15] Further activation of the remaining pendant alkyne moiety by the gold(I) complex would result in either a step-wise or concerted formal [4+2] cycloaddition[16,17] to deliver the Wheland intermediate **VIIa**, which upon rearomatization followed by protondeauration, yields the final product **2a** and regenerates the gold(I) catalyst.



Scheme 5. Proposed mechanism for gold(I)-catalyzed sequential cycloisomerization of 1,3,9-ene-allene-ynes represented by **1a**.

Conclusion

In summary, we have developed an efficient strategy for the synthesis of spiro tetrahydro-*3H*-dicyclopenta[*a,b*] naphthalenes from gold catalyzed sequential cycloisomerization of 1,3,9-ene-allene-ynes. The transformation has excellent functional group tolerance and provided the corresponding products in good to excellent yields under mild conditions. The synthetic utility of the present method was also demonstrated by the scale-up preparation of one example followed by its conversion to the corresponding dioxides and diketones. Efforts to explore the synthetic applications of the present reactions are currently underway and will be reported in due course.

Experimental Section

To a solution of 1,3,9-ene-allene-yne **1** (0.3 mmol) in DCE (3 mL) was added gold(I) complex **C** (14 mg, 15 μmol, 5 mol-%) and the reaction mixture was stirred at room temperature for 12 h. Upon completion, the solvent was removed under reduced pressure and the residue was purified by flash column chromatography on silica gel (eluent: petroleum ether/Et_2O = 100:1 to 10:1) to give the desired product **2** in 60–96 % yields.

2-(3-Tosylpropyl)-4,5,6,11b-tetrahydro-3*H*-dicyclopenta-[*a,b*]naphthaalene (2a): Colorless solid (109 mg, 90 % yield), m.p. 96–98 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.70 (d, *J* = 8.2 Hz, 2H), 7.33 (d, *J* = 8.1 Hz, 2H), 7.09–7.06 (m, 2H), 7.04–7.01 (m, 1H), 7.00–6.96 (m, 1H), 6.09 (s, 1H), 5.90 (s, 1H), 3.70 (s, 1H), 3.06–2.94 (m, 2H), 2.61–2.52 (m, 1H), 2.52–2.47 (m, 1H), 2.45 (s, 3H), 2.14–2.10 (m, 2H), 2.04 (d, *J* = 4.0 Hz, 2H), 1.91–1.70 (m, 5H), 1.53–1.45 (m, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 148.4, 144.5, 142.5, 136.2, 136.1, 133.4, 129.9, 128.2, 128.0, 127.2, 126.5, 126.1, 126.0, 117.3, 55.5, 52.4, 52.4, 47.4, 41.5, 29.5, 29.5, 21.7, 20.5, 20.5. HRMS (ESI) calcd. for $C_{26}H_{29}O_2S$ $[M + H]^+$: 405.1883, found 405.1881.

Ethyl 4-(4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a,b*]naphthalen-2-yl)butanoate (2b): Colorless oil (81 mg, 84 % yield); ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.11–7.06 (m, 3H), 6.99–6.95 (m, 1H), 6.10 (s, 1H), 5.95 (s, 1H), 4.13 (q, *J* = 7.1 Hz, 2H), 3.74 (s, 1H), 2.65–2.57 (m, 1H), 2.53–2.46 (m, 1H), 2.28 (td, *J* = 7.6, 2.9 Hz, 2H), 2.26–2.14 (m, 2H), 2.10 (t, *J* = 7.5 Hz, 2H), 1.93–1.88 (m, 1H), 1.86–1.73 (m, 4H), 1.53 (dd, *J* = 20.7, 10.5 Hz, 1H), 1.26 (t, *J* = 7.1 Hz, 3H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 173.6, 148.7, 143.9, 136.4, 133.4, 127.3, 126.9, 126.4, 126.0, 125.9, 117.2, 60.2, 52.5, 52.4, 47.9, 41.6, 33.9, 30.6, 29.6, 22.9, 20.5, 14.3. HRMS (ESI) calcd. for $C_{22}H_{27}O_2$ $[M + H]^+$: 323.2006, found 323.2015.

3-(4,5,6,11b-Tetrahydro-3*H*-dicyclopenta[*a,b*]naphthalen-2-yl)-propyl 4-Nitrobenzoate (2c): Colorless oil (118 mg, 95 % yield); ^{1}H NMR (600 MHz, $CDCl_3$) δ = 8.28 (d, *J* = 8.9 Hz, 2H), 8.20 (d, *J* = 8.9 Hz, 2H), 7.10–7.04 (m, 3H), 6.97 (dd, *J* = 6.5, 2.5 Hz, 1H), 6.10 (s, 1H), 5.98 (s, 1H), 4.41–4.31 (m, 2H), 3.76 (s, 1H), 2.65–2.57 (m, 1H), 2.54–2.45 (m, 1H), 2.29–2.18 (m, 4H), 1.99–1.88 (m, 3H), 1.86–1.77 (m, 2H), 1.59–1.49 (m, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 164.7, 150.5, 148.5, 143.4, 136.2, 135.8, 133.4, 130.7, 127.2, 127.1, 126.4, 126.1, 126.0, 123.5, 117.3, 65.6, 52.5, 52.4, 48.1, 41.5, 29.5, 27.7, 26.6, 20.4. HRMS (ESI) calcd. for $C_{26}H_{25}NO_4Na$ $[M + Na]^+$: 438.1676, found 438.1684.

2-(3-Phenoxypropyl)-4,5,6,11b-tetrahydro-3*H*-dicyclopenta-[*a,b*]naphthalene (2d): Colorless solid (90 mg, 88 % yield), m.p. 115–116 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.33–7.27 (m, 2H), 7.12–7.10 (m, 3H), 7.01–7.00 (m, 1H), 6.97 (t, *J* = 7.3 Hz, 1H), 6.89 (d, *J* = 7.9 Hz, 2H), 6.14 (s, 1H), 6.00 (s, 1H), 3.95 (td, *J* = 6.5, 1.9 Hz, 2H), 3.78 (s, 1H), 2.68–2.60 (m, 1H), 2.57–2.49 (m, 1H), 2.32–2.21 (m, 4H), 2.01–1.92 (m, 3H), 1.91–1.81 (m, 2H), 1.60–1.52 (m, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 159.0, 148.7, 144.0, 136.4, 133.4, 129.4, 127.3, 126.8, 126.4, 126.0, 125.9, 120.5, 117.3, 114.5, 67.3, 52.5, 48.0, 41.6, 29.6, 27.7, 27.2, 20.5. HRMS (ESI) calcd. for $C_{25}H_{27}O$ $[M + H]^+$: 343.2056, found 343.2063.

4-Methyl-*N*-phenyl-*N*-(3-(4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a,b*]-naphthalen-2-yl)propyl)benzenesulfonamide (2e): Colorless solid (137 mg, 92 % yield), m.p. 107–108 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.45 (d, *J* = 8.1 Hz, 2H), 7.32–7.28 (m, 3H), 7.24 (d, *J* = 8.0 Hz, 2H), 7.10–6.96 (m, 7H), 6.10 (s, 1H), 5.88 (s, 1H), 3.70 (s, 1H), 3.58–3.51 (m, 1H), 3.48–3.41 (m, 1H), 2.62–2.55 (m, 1H), 2.49 (dd, *J* = 17.3, 8.6 Hz, 1H), 2.43 (s, 3H), 2.16–2.04 (m, 4H), 1.88–1.75 (m, 3H), 1.66–1.56 (m, 2H), 1.50 (dd, *J* = 20.6, 10.3 Hz, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 148.7, 143.5, 143.3, 139.4, 136.4, 135.4, 133.5, 129.4, 129.0, 128.8, 127.8, 127.7, 127.3, 127.0, 126.4, 126.0, 125.9, 117.2, 52.5, 52.4, 50.3, 47.8, 41.5, 29.6, 28.0, 26.3, 21.6, 20.5. HRMS (ESI) calcd. for $C_{32}H_{34}NO_2S$ $[M + H]^+$: 496.2305, found 496.2316.

2-(3-(4,5,6,11b-Tetrahydro-3*H*-dicyclopenta[*a,b*]naphthalen-2-yl)propyl)isoindoline-1,3-dione (2f): Colorless solid (114 mg, 96 % yield), m.p. 92–93 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.84 (dd, *J* = 5.4, 3.0 Hz, 2H), 7.70 (dd, *J* = 5.4, 3.0 Hz, 2H), 7.10–7.03 (m, 3H),

 © 2020 Wiley-VCH GmbH

6.96–6.93 (m, 1H), 6.08 (s, 1H), 5.98 (s, 1H), 3.72 (s, 1H), 3.71–3.64 (m, 2H), 2.63–2.55 (m, 1H), 2.51–2.45 (m, 1H), 2.24–2.17 m, 2H), 2.12 (t, *J* = 7.6 Hz, 2H), 1.95–1.78 (m, 5H), 1.51 (dd, *J* = 20.2, 11.1 Hz, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 168.4, 148.7, 143.5, 136.3, 133.9, 133.4, 132.2, 127.3, 126.8, 126.4, 126.0, 125.9, 123.2, 117.2, 52.5, 52.4, 48.0, 41.5, 37.8, 29.5, 28.6, 26.5, 20.4. HRMS (ESI) calcd. for $C_{27}H_{26}NO_2$ $[M + H]^+$: 396.1958, found 396.1973.

2-Methyl-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalene (2g): Colorless solid (57 mg, 85 % yield), m.p. 59–60 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.12–7.05 (m, 3H), 6.98 (dd, *J* = 5.7, 2.1 Hz, 1H), 6.11 (t, *J* = 1.8 Hz, 1H), 5.91–5.89 (m, 1H), 3.76 (s, 1H), 2.66–2.58 (m, 1H), 2.55–2.47 (m, 1H), 2.30–2.26 (m, 1H), 2.16 (d, *J* = 15.6 Hz, 1H), 1.97–1.91 (m, 1H), 1.86–1.80 (m, 2H), 1.74 (s, 3H), 1.55 (dd, *J* = 20.7, 11.2 Hz, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 148.8, 140.6, 136.6, 133.4, 127.3, 127.2, 126.4, 126.0, 126.0, 117.1, 52.8, 52.8, 50.0, 41.7, 29.6, 20.4, 17.0. HRMS (ESI) calcd. for $C_{17}H_{19}$ $[M + H]^+$: 223.1481, found 223.1490.

2-Decyl-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalene (2h): Colorless solid (94 mg, 90 % yield), m.p. 52–54 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.16–7.03 (m, 3H), 7.03–6.88 (m, 1H), 6.11 (s, 1H), 5.91 (s, 1H), 3.74 (s, 1H), 2.69–2.57 (m, 1H), 2.57–2.44 (m, 1H), 2.20 (dd, *J* = 38.1, 15.5 Hz, 2H), 2.05 (s, 2H), 1.98–1.88 (m, 1H), 1.88–1.77 (m, 2H), 1.53 (dd, *J* = 20.5, 10.6 Hz, 1H), 1.43 (d, *J* = 6.2 Hz, 2H), 1.27 (s, 14H), 0.91 (td, *J* = 7.0, 1.1 Hz, 3H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 148.9, 145.2, 136.6, 133.4, 127.3, 126.4, 125.9, 125.9, 125.8, 117.2, 52.5, 52.5, 48.0, 41.6, 32.0, 31.3, 29.6, 29.6, 29.5, 29.5, 29.4, 27.6, 22.7, 20.5, 14.2. HRMS (ESI) calcd. for $C_{26}H_{37}$ $[M + H]^+$: 349.2890, found 349.2907.

2-(3-Phenylpropyl)-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalene (2i): Colorless oil (79 mg, 81 % yield); ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.30 (t, *J* = 7.5 Hz, 2H), 7.20 (dd, *J* = 18.7, 7.4 Hz, 3H), 7.14–7.08 (m, 3H), 7.01–6.98 (m, 1H), 6.13 (s, 1H), 5.96 (s, 1H), 3.77 (s, 1H), 2.63–2.49 (m, 3H), 2.57–2.48 (m, 1H), 2.23 (dd, *J* = 38.6, 15.5 Hz, 2H), 2.12 (t, *J* = 7.5 Hz, 2H), 1.97–1.91 (m, 1H), 1.88–1.75 (m, 4H), 1.60–1.51 (m, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 148.8, 144.6, 142.5, 136.5, 133.5, 128.5,128.3, 127.3, 126.4, 126.0, 126.0, 125.7, 117.2, 52.5, 52.5, 48.1, 41.6, 35.7, 30.9, 29.6, 29.4, 20.5. HRMS (ESI) calcd. for $C_{25}H_{27}$ $[M + H]^+$: 327.2107, found 327.2110.

2-Cyclohexyl-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]-naphthalene (2j): Colorless solid (71 mg, 82 % yield), m.p. 82–83 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.12–7.03 (m, 3H), 7.00–6.94 (m, 1H), 6.11 (s, 1H), 5.89 (d, *J* = 1.3 Hz, 1H), 3.71 (d, *J* = 2.2 Hz, 1H), 2.65–2.56 (m, 1H), 2.54–2.46 (m, 1H), 2.21 (s, 2H), 1.97 (t, *J* = 11.3 Hz, 1H), 1.92–1.86 (m, 1H), 1.86–1.78 (m, 3H), 1.77–1.70 (m, 3H), 1.67 (d, *J* = 12.5 Hz, 1H), 1.56–1.47 (m, 1H), 1.32–1.22 (m, 2H), 1.20–1.09 (m, 3H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 150.2, 148.9, 136.7, 133.5, 127.3, 126.3, 125.8, 125.8, 123.8, 117.2, 52.2, 52.2, 46.1, 41.5, 39.7, 31.8, 31.7, 29.6, 27.0, 26.4, 26.4, 20.5. HRMS (ESI) calcd. for $C_{22}H_{27}$ $[M + H]^+$: 291.2107, found 291.2122.

4-Bromo-*N*-(2-decyl-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalen-10-yl)-*N*-methylbenzenesulfonamide (2k): Colorless solid (147 mg, 82 % yield), m.p. 70–71 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.58 (d, *J* = 8.6 Hz, 2H), 7.42 (d, *J* = 8.6 Hz, 2H), 6.87 (d, *J* = 8.1 Hz, 1H), 6.79 (s, 1H), 6.73 (dd, *J* = 8.0, 2.0 Hz, 1H), 6.05 (s, 1H), 5.70 (s, 1H), 3.66 (s, 1H), 3.13 (s, 3H), 2.65–2.55 (m, 1H), 2.51–2.45 (m, 1H), 2.18 (d, *J* = 3.5 Hz, 1H), 2.11–2.00 (m, 2H), 1.91–1.85 (m, 1H), 1.85–1.77 (m, 2H), 1.50 (dd, *J* = 20.6, 10.5 Hz, 1H), 1.45–1.37 (m, 2H), 1.30–1.25 (m, 15H), 0.88 (t, *J* = 7.0 Hz, 3H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 150.2, 145.6, 139.0, 137.5, 135.8, 133.0, 131.9, 129.5, 127.6, 126.1, 125.9, 125.4, 124.1, 116.3, 52.4, 52.3, 48.1, 41.5, 38.3, 31.9, 31.3, 29.6, 29.6, 29.5, 29.5, 29.4, 27.7, 22.7, 20.4, 14.2. HRMS (ESI) calcd. for $C_{33}H_{42}BrNO_2SNa$ $[M + Na]^+$: 618.2012, found 618.2045.

4-Bromo-*N*-(2-(3-((4-fluorophenyl)sulfonyl)propyl)-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalen-10-yl)-*N*-methylbenzenesulfonamide (2l): Colorless solid (138 mg, 70 % yield), m.p. 73–75 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.87–7.84 (m, 2H), 7.58 (d, *J* = 8.6 Hz, 2H), 7.41 (d, *J* = 8.6 Hz, 2H), 7.21 (t, *J* = 8.5 Hz, 2H), 6.87 (d, *J* = 7.8 Hz, 2H), 6.63 (dd, *J* = 8.0, 1.8 Hz, 1H), 6.06 (s, 1H), 5.78 (s, 1H), 3.67 (s, 1H), 3.13 (s, 3H), 3.08–3.03 (m, 1H), 3.02–2.94 (m, 1H), 2.60–2.55 (m, 1H), 2.51–2.44 (m, 1H), 2.17 (t, *J* = 7.1 Hz, 2H), 2.07 (s, 2H), 1.89–1.71 (m, 5H), 1.54–1.46 (m, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 165.8 (d, *J* = 255.0 Hz, 1C), 149.8, 142.8, 139.1, 137.0, 135.7, 132.9, 132.0, 130.8 (d, *J* = 8.1 Hz, 1C), 129.5, 127.9, 127.7, 126.6, 126.1, 123.5, 116.7, 116.5 (d, *J* = 15.2 Hz, 1C), 116.5, 55.5, 52.3, 52.2, 47.5, 41.4, 38.3, 29.5, 29.5, 20.6, 20.4. HRMS (ESI) calcd. for $C_{32}H_{31}BrFNO_4S_2Na$ $[M + Na]^+$: 678.0754, found 678.0779.

4-Bromo-*N*-(2-(3-((4-chlorophenyl)sulfonyl)propyl)-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalen-10-yl)-*N*-methylbenzenesulfonamide (2m): Colorless solid (143 mg, 71 % yield), m.p. 82–83 °C; 1H NMR (600 MHz, CDCl3) δ = 7.76 (d, *J* = 8.6 Hz, 2H), 7.58 (d, *J* = 8.6 Hz, 2H), 7.50 (d, *J* = 8.6 Hz, 2H), 7.41 (d, *J* = 8.6 Hz, 2H), 6.87 (d, *J* = 8.2 Hz, 2H), 6.63 (dd, *J* = 8.0, 1.8 Hz, 1H), 6.06 (s, 1H), 5.78 (s, 1H), 3.67 (s, 1H), 3.12 (s, 3H), 3.07–3.02 (m, 1H), 3.00–2.95 (m, 1H), 2.62–2.52 (m, 1H), 2.50–2.44 (m, 1H), 2.16 (t, *J* = 7.1 Hz, 2H), 2.05 (s, 2H), 1.90–1.72 (m, 5H), 1.54–1.44 (m, 1H); ^{13}C NMR (150 MHz, CDCl3) δ = 149.8, 142.7, 140.4, 139.1, 137.5, 137.0, 135.7, 132.9, 132.0, 129.7, 129.5, 129.5, 128.0, 127.7, 126.5, 126.1, 123.5, 116.5, 55.4, 52.3, 52.2, 47.5, 41.4, 38.3, 29.5, 29.4, 20.5, 20.4. HRMS (ESI) calcd. for C32H31BrClNO4S2Na [M + Na]+: 694.0459, found 694.0492.

10-(4-Bromophenyl)-2-(3-tosylpropyl)-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalene (2n): Colorless solid (134 mg, 80 % yield), m.p. 134–135 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.71 (d, *J* = 8.2 Hz, 2H), 7.56 (d, *J* = 8.5 Hz, 2H), 7.43 (d, *J* = 8.5 Hz, 2H), 7.29 (d, *J* = 9.4 Hz, 3H), 7.24 (s, 1H), 7.06 (d, *J* = 7.8 Hz, 1H), 6.16 (s, 1H), 5.99 (s, 1H), 5.78 (s, 1H), 3.10–2.96 (m, 2H), 2.66–2.58 (m, 1H), 2.58–2.49 (m, 1H), 2.43 (s, 3H), 2.22–2.06 (m, 4H), 1.96–1.75 (m, 5H), 1.59–1.51 (m, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 149.2, 144.6, 142.9, 140.2, 138.0, 136.8, 136.2, 133.1, 131.8, 129.9, 128.5, 128.0, 127.9, 126.5, 125.8, 124.7, 121.1, 116.9, 55.6, 52.6, 52.5, 47.5, 41.5, 29.6, 29.5, 21.6, 20.5, 20.4. HRMS (ESI) calcd. for $C_{32}H_{32}BrO_2S$ $[M + H]^+$: 559.1301, found 559.1319.

2-Methyl-10-phenyl-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalene (2o): Colorless solid (74 mg, 83 % yield), m.p. 122–124 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.60 (d, *J* = 7.2 Hz, 2H), 7.43 (t, *J* = 7.7 Hz, 2H), 7.37–7.30 (m, 3H), 7.06 (d, *J* = 7.6 Hz, 1H), 6.15 (s, 1H), 5.96 (s, 1H), 3.81 (s, 1H), 2.68–2.61 (m, 1H), 2.57–2.49 (m, 1H), 2.32 (dd, *J* = 15.6, 1.1 Hz, 1H), 2.18 (d, *J* = 15.6 Hz, 1H), 1.99–1.94 (m, 1H), 1.87–1.82 (m, 2H), 1.75 (s, 3H), 1.57 (dd, *J* = 20.4, 10.9 Hz, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 149.2, 141.5, 140.7, 139.3, 137.0, 132.7, 128.7, 127.1, 126.9, 126.9, 126.3, 126.3, 124.8, 116.8, 52.9, 50.1, 41.7, 29.7, 29.6, 20.4, 17.0. HRMS (ESI) calcd. for $C_{23}H_{23}$ $[M + H]^+$: 299.1794, found 299.1806.

2-Methyl-10-(trifluoromethyl)-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalene (2p): Colorless oil (54 mg, 62 % yield); ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.30 (d, *J* = 8.5 Hz, 2H), 7.03 (d, *J* = 7.7 Hz, 1H), 6.12 (s, 1H), 5.89 (s, 1H), 3.73 (s, 1H), 2.67–2.58 (m, 1H), 2.57–2.46 (m, 1H), 2.24 (dd, *J* = 15.7, 1.2 Hz, 1H), 2.15 (d, *J* = 15.7 Hz, 1H), 1.97–1.91 (m, 1H), 1.87–1.79 (m, 2H), 1.74 (s, 3H), 1.55 (s, 1H). ^{13}C NMR (150 MHz, $CDCl_3$) δ = 152.1, 141.6, 137.1, 136.7, 128.0 (q, *J* = 31.9 Hz, 1C), 126.3, 125.8, 125.4 (q, *J* = 270 Hz, 1C), 124.2 (q, *J* =

© 2020 Wiley-VCH GmbH

3.7 Hz, 1C), 122.9 (q, *J* = 3.8 Hz, 1C), 116.4, 52.9, 52.5, 49.9, 41.5, 29.7, 20.3, 17.0. HRMS (ESI) calcd. for $C_{18}H_{18}F_3$ [M + H]$^+$: 291.1355, found 291.1364.

2,8-Dimethyl-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]-naphthalene (2q): Colorless solid (55 mg, 78 % yield), m.p. 53–54 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.03–6.90 (m, 3H), 6.32 (s, 1H), 5.88 (s, 1H), 3.74 (s, 1H), 2.68–2.62 (m, 1H), 2.58–2.48 (m, 1H), 2.33 (s, 3H), 2.28 (dd, *J* = 15.6, 0.9 Hz, 1H), 2.14 (d, *J* = 15.6 Hz, 1H), 1.97–1.90 (m, 1H), 1.85–1.78 (m, 2H), 1.73 (s, 3H), 1.57–1.48 (m, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 149.0, 140.5, 136.6, 132.6, 131.5, 127.8, 127.5, 125.9, 125.1, 113.8, 53.2, 52.2, 50.0, 42.0, 30.1, 20.5, 19.8, 17.0. HRMS (ESI) calcd. for $C_{18}H_{21}$ [M + H]$^+$: 237.1638, found 237.1647.

2,9-Dimethyl-4,5,6,11b-tetrahydro-3H-dicyclopenta[a,b]-naphthalene (2r): an inseparable mixture of regioisomers in a ratio of 10:1, Colrless oil (55 mg, 77 % yield); major isomer: ^{1}H NMR (400 MHz, $CDCl_3$) δ = 7.00 (d, *J* = 7.7 Hz, 1H), 6.93 (d, *J* = 7.7 Hz, 1H), 6.82 (s, 1H), 6.08 (s, 1H), 5.90 (s, 1H), 3.72 (s, 1H), 2.66–2.46 (m, 2H), 2.36–2.26 (m, 4H), 2.15 (d, *J* = 15.5 Hz, 1H), 1.95–1.80 (m, 2H), 1.74 (s, 3H), 1.58–1.51 (m, 1H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 148.9, 140.4, 135.3, 133.6, 133.3, 127.3, 127.2, 127.1, 126.7, 117.1, 52.9, 52.5, 50.0, 41.6, 29.6, 21.0, 20.5, 17.0. HRMS (ESI) calcd. for $C_{18}H_{21}$ [M + H]$^+$: 237.1638, found 237.1643.

2-Methyl-4,5,6,13b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]phenanthrene (2s): Colorless oil (60 mg, 74 % yield); ^{1}H NMR (600 MHz, $CDCl_3$) δ = 8.13 (d, *J* = 8.5 Hz, 1H), 7.76 (d, *J* = 8.1 Hz, 1H), 7.62 (d, *J* = 8.4 Hz, 1H), 7.46 (t, *J* = 7.4 Hz, 1H), 7.42–7.37 (m, 1H), 7.29 (d, *J* = 8.4 Hz, 1H), 6.92 (s, 1H), 5.97 (s, 1H), 3.93 (s, 1H), 2.79–2.72 (m, 1H), 2.68–2.59 (m, 1H), 2.33 (d, *J* = 15.6 Hz, 1H), 2.19 (d, *J* = 15.6 Hz, 1H), 2.02–1.95 (m, 1H), 1.91–1.83 (m, 2H), 1.74 (s, 3H), 1.65–1.57 (m, 1H). ^{13}C NMR (150 MHz, $CDCl_3$) δ = 149.9, 141.6, 133.3, 132.5, 129.5, 128.3, 127.9, 127.1, 126.2, 126.1, 125.6, 124.7, 123.3, 112.4, 53.8, 52.5, 50.3, 42.0, 30.3, 20.7, 17.0. HRMS (ESI) calcd. for $C_{21}H_{21}$ [M + H]$^+$: 273.1638, found 273.1643.

7-Methyl-3-tosyl-3,5b,8,9,10,11-hexahydrocyclopenta[7,7*a*]-indeno[5,4-*e*]indole (2t): Colorless oil (95 mg, 76 % yield); ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.75 (d, *J* = 8.4 Hz, 2H), 7.71 (d, *J* = 8.5 Hz, 1H), 7.51 (d, *J* = 3.7 Hz, 1H), 7.18 (d, *J* = 8.2 Hz, 2H), 7.06 (d, *J* = 8.5 Hz, 1H), 6.71 (d, *J* = 3.6 Hz, 1H), 6.38 (s, 1H), 5.90 (s, 1H), 3.79 (s, 1H), 2.70–2.61 (m, 1H), 2.59–2.47 (m, 1H), 2.31 (s, 3H), 2.25 (d, *J* = 15.4 Hz, 1H), 2.15 (d, *J* = 15.6 Hz, 1H), 1.97–1.89 (m, 1H), 1.84–1.80 (m, 2H), 1.71 (s, 3H), 1.53 (dd, *J* = 20.2, 11.0 Hz, 1H). ^{13}C NMR (150 MHz, $CDCl_3$) δ = 150.4, 144.7, 140.7, 135.5, 133.5, 130.6, 129.8, 127.5, 127.2, 126.9, 125.8, 125.7, 124.3, 112.9, 111.2, 107.1, 52.9, 52.6, 50.2, 41.7, 29.8, 21.5, 20.4, 17.0. HRMS (ESI) calcd. for $C_{26}H_{26}NO_2S$ [M + H]$^+$: 416.1679, found 416.1690.

2-Methyl-4,5,6,10b-tetrahydro-3*H*-cyclopenta[7,7*a*]indeno-[4,5-*b*]thiophene (2u): Colorless solid (45 mg, 66 % yield), m.p. 57–58 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.12 (d, *J* = 5.2 Hz, 1H), 6.95 (d, *J* = 5.2 Hz, 1H), 6.09 (t, *J* = 3.3 Hz, 1H), 5.91 (d, *J* = 5.2 Hz, 1H), 5.76 (d, *J* = 5.2 Hz, 1H), 2.30–2.17 (m, 2H), 1.85–1.83 (m, 4H), 1.77–1.70 (m, 1H), 1.69–1.62 (m, 1H), 1.56 (s, 3H). ^{13}C NMR (150 MHz, $CDCl_3$) δ = 145.1, 141.5, 134.3, 133.6, 131.8, 123.5, 120.3, 120.2, 57.9, 49.9, 47.5, 32.3, 25.8, 21.6, 20.0. HRMS (ESI) calcd. for $C_{15}H_{17}S$ [M + H]$^+$: 229.1045, found 229.1053.

2,3-Dibutyl-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]naphthalene (2v): an inseparable mixture of diastereoisomers in a ratio of 2.6:1, colorless solid (61 mg, 63 % yield), m.p. 51–53 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.11–7.02 (m, 3H), 7.00–6.97 (m, 0.72H), 6.95–6.92 (m, 0.28H), 6.28 (s, 0.28H), 6.19 (s, 0.72H), 5.95 (s, 0.28H), 5.89 (s, 0.72H), 3.63 (s, 0.28H), 3.51 (s, 0.72H), 2.72–2.62 (m, 0.8H), 2.60–2.43 (m, 2.2H), 2.19–2.13 (m, 1H), 2.05–2.01 (m, 1H), 1.97–1.89 (m, 0.83H), 1.80–1.72 (m, 2.24H), 1.56–1.28 (m, 11H), 0.94–0.87 (m, 6H); ^{13}C NMR (150 MHz, $CDCl_3$) δ = 150.1, 149.8, 149.0, 146.7, 138.1, 136.6, 133.8, 127.1, 126.3, 126.2, 126.2, 125.9, 125.7, 125.6, 125.5, 125.4, 124.7, 120.6, 118.1, 57.7, 57.1, 55.8, 54.8, 53.1, 52.2, 43.6, 36.2, 32.0, 31.9, 31.4, 31.0, 30.6, 30.0, 30.0, 29.9, 29.7, 29.4, 27.4, 23.4, 22.9, 22.7, 22.7, 21.9, 20.7, 14.1, 14.1, 14.0. HRMS (ESI) calcd. for $C_{24}H_{33}$ [M + H]$^+$: 321.2577, found 321.2507.

2,3,4,4a,5,6,7,12b-Octahydro-1*H*-benzo[*a*]cyclopenta[*d*]fluorene (2w): an inseparable mixture of diastereoisomers in a ratio of 6.1:1, colorless solid (47 mg, 60 % yield), m.p. 104–105 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.10–7.03 (m, 3H), 7.00–6.95 (m, 0.86H), 6.95–6.92 (m, 0.14H), 6.20 (s, 0.14H), 6.09 (s, 0.86H), 5.82 (dd, *J* = 5.2, 2.6 Hz, 0.86H), 5.75 (s, 0.14H), 3.85 (s, 0.14H), 3.67 (s, 0.86H), 2.65–2.46 (m, 3H), 2.44 (dd, *J* = 12.6, 4.4 Hz, 0.17H), 2.39 (dd, *J* = 12.2, 5.1 Hz, 0.86H), 2.27 (ddd, *J* = 12.9, 8.4, 1.6 Hz, 0.93H), 2.09 (dd, *J* = 12.4, 4.6 Hz, 0.14H), 1.95–1.88 (m, 1H), 1.83–1.65 (m, 5H), 1.48 (dt, *J* = 13.0, 10.3 Hz, 1H), 1.44–1.35 (m, 1H), 1.24–1.13 (m, 2H). ^{13}C NMR (150 MHz, $CDCl_3$) δ = 149.8, 148.1, 145.6, 144.9, 137.5, 136.6, 133.2, 127.2, 126.8, 126.4, 126.2, 126.0, 125.9, 125.8, 125.7, 124.0, 121.8, 119.1, 116.7, 58.4, 54.9, 54.5, 52.8, 52.7, 43.2, 37.2, 32.1, 30.5, 29.9, 29.9, 28.9, 28.7, 28.0, 26.3, 25.8, 25.5, 20.4, 19.9. HRMS (ESI) calcd. for $C_{20}H_{23}$ [M + H]$^+$: 263.1794, found 263.1804.

1,2-Dimethyl-4,5,6,11b-tetrahydro-3*H*-dicyclopenta[*a*,*b*]-naphthalene (2x): Colorless solid (45 mg, 64 % yield), m.p. 54–56 °C; ^{1}H NMR (600 MHz, $CDCl_3$) δ = 7.34 (d, *J* = 7.2 Hz, 1H), 7.12–7.06 (m, 2H), 7.01–6.98 (m, 1H), 6.11 (s, 1H), 3.57 (s, 1H), 2.66–2.59 (m, 1H), 2.53–2.45 (m, 1H), 2.34 (dd, *J* = 15.1, 1.0 Hz, 1H), 2.11 (d, *J* = 15.2 Hz, 1H), 2.06 (s, 3H), 1.90–1.85 (m, 1H), 1.83–1.77 (m, 2H), 1.63 (s, 3H), 1.55–1.47 (m, 1H). ^{13}C NMR (150 MHz, $CDCl_3$) δ = 149.2, 136.2, 134.2, 133.5, 132.9, 126.9, 126.1, 126.1, 126.0, 117.1, 57.7, 52.5, 50.6, 41.0, 29.6, 20.5, 16.1, 14.5. HRMS (ESI) calcd. for $C_{18}H_{21}$ [M + H]$^+$: 237.1638, found 237.1643.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 21502093), Project Funds from the National First-Class Disciplines (PNFD) and Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD).

Keywords: Fused carbocycles · Gold · Sequential cycloisomerization · Ene-allene-ynes · Synthetic methods

[1] Selected reviews, see: a) P.-W. Xu, J.-S. Yu, C. Chen, Z.-Y. Cao, F. Zhou, J. Zhou, *ACS Catal.* **2019**, *9*, 1820–1882; b) T. P. Stockdale, C. M. Williams, *Chem. Soc. Rev.* **2015**, *44*, 7737–7763; c) L. K. Smith, I. R. Baxendale, *Org. Biomol. Chem.* **2015**, *13*, 9907–9933; d) V. A. D'yakonov, O. A. Trapeznikova, A. de Meijere, U. M. Dzhemilev, *Chem. Rev.* **2014**, *114*, 5775–5814; e) R. Rios, *Chem. Soc. Rev.* **2012**, *41*, 1060–1074; f) G. Mehta, A. Srikrishna, *Chem. Rev.* **1997**, *97*, 671–719.

[2] T. Roncal, S. Cordobés, U. Ugalde, Y. He, O. Sterner, *Tetrahedron Lett.* **2002**, *43*, 6799–6802.

[3] S. Niu, Z.-W. Fan, C.-L. Xie, Q. Liu, Z.-H. Luo, G. Liu, X.-W. Yang, *J. Nat. Prod.* **2017**, *80*, 2174–2177.

[4] a) H. Lee, T. Kang, H.-Y. Lee, *Angew. Chem. Int. Ed.* **2017**, *56*, 8254; *Angew. Chem.* **2017**, *129*, 8366; b) D. B. Clarke, S. F. R. Hinkley, R. T. Weavers, *Tetrahedron Lett.* **1997**, *38*, 4297–4300.

[5] T. Toyomasu, A. Kaneko, T. Tokiwano, Y. Kanno, Y. Kanno, R. Niida, S. Miura, T. Nishioka, C. Ikeda, W. Mitsuhashi, T. Dairi, T. Kawano, H. Oikawa, N. Kato, T. Sassa, *J. Org. Chem.* **2009**, *74*, 1541–1548.

© 2020 Wiley-VCH GmbH

[6] X.-J. Wang, G. J. Zhang, P.-Y. Zhuang, Y. Zhang, S.-S. Yu, X.-Q. Bao, D. Zhang, Y.-H. Yuan, N.-H. Chen, S.-G. Ma, J. Qu, Y. Li, *Org. Lett.* **2012**, *14*, 2614–2617.

[7] J.-P. Wang, Y. Shu, J.-T. Hu, R. Liu, X.-Y. Cai, C.-T. Sun, D. Gan, D.-J. Zhou, R.-F. Mei, H. Ding, X.-R. Zhang, L. Cai, Z.-T. Ding, *Org. Chem. Front.* **2020**, *7*, DOI:-10.1039/d0qo00301h.

[8] a) Y.-J. Kim, Y. Yoon, H.-Y. Lee, *Tetrahedron* **2013**, *69*, 7810–7816; b) S. Takamatsu, L. H. Xu, S. Fushinobu, H. Shoun, M. Komatsu, D. E. Cane, H. Ikeda, *J. Antibiot.* **2011**, *64*, 65–71; c) P. Joseph-Nathan, B. Reyes-Trejo, M. S. Morales-Ríos, *J. Org. Chem.* **2006**, *71*, 4411–4417.

[9] Selected reviews on gold catalysis, see: a) Y. Wei, M. Shi, *ACS Catal.* **2016**, *6*, 2515–2524; b) D. P. Day, P. W. H. Chan, *Adv. Synth. Catal.* **2016**, *358*, 1368–1384; c) R. Dorel, A. M. Echavarren, *Chem. Rev.* **2015**, *115*, 9028–9072; d) *Gold catalysis: a homogeneous approach* (Eds.: F. D. Toste, V. Michelet), Imperial College Press, London, **2014**; e) A. S. K. Hashmi, *Acc. Chem. Res.* **2014**, *47*, 864–876; f) *Modern gold catalyzed synthesis* (Eds.: A. S. K. Hashmi, F. D. Toste), Wiley-VCH, Weinheim, **2012**; g) A. Fürstner, *Chem. Soc. Rev.* **2009**, *38*, 3208–3221; h) A. S. K. Hashmi, *Chem. Rev.* **2007**, *107*, 3180–3211.

[10] Selected reviews on gold-catalyzed cycloisomerization of 1,n-enynes, see: a) C. Obradors, A. M. Echavarren, *Acc. Chem. Res.* **2014**, *47*, 902–912; b) V. Michelet, P. Y. Toullec, J. P. Genet, *Angew. Chem. Int. Ed.* **2008**, *47*, 4268–4315; *Angew. Chem.* **2008**, *120*, 4338; Selected recent examples: c) Y.-C. Lee, S. Patil, C. Golz, C. Strohmann, S. Ziegler, K. Kumar, H. Waldmann, *Nat. Commun.* **2017**, *8*, 14043–14054; d) G.-Q. Chen, W. Fang, Y. Wei, X.-Y. Tang, M. Shi, *Chem. Sci.* **2016**, *7*, 4318–4328; e) H. Zheng, L. L. Adduci, R. J. Felix, M. R. Gagné, *Angew. Chem. Int. Ed.* **2014**, *53*, 7904–7997; *Angew. Chem.* **2014**, *126*, 8038; f) A. K. Buzas, F. M. Istrate, F. Gagosz, *Angew. Chem. Int. Ed.* **2007**, *46*, 1141–1144; *Angew. Chem.* **2007**, *119*, 1159.

[11] For reviews on cycloisomerization of 1,n-allenenes and 1,n-allenynes: a) C. Aubert, L. Fensterbank, P. Garcia, M. Malacria, A. Simonneau, *Chem. Rev.* **2011**, *111*, 1954–1993; Selected examples: b) F. Jaroschik, A. Simonneau, G. Lemière, K. Cariou, N. Agenet, H. Amouri, C. Aubert, J.-P. Goddard, D. Lesage, M. Malacria, Y. Gimbert, V. Gandon, L. Fensterbank, *ACS Catal.* **2016**, *6*, 5146–5160; c) L.-Y. Mei, Y. Wei, X.-Y. Tang, M. Shi, *J. Am. Chem. Soc.* **2015**, *137*, 8131–8137; d) A. Pitaval, D. Leboeuf, J. Ceccon, A. M. Echavarren, *Org. Lett.* **2013**, *15*, 4580–4583; e) P. H.-Y. Cheong, P. Morganelli, M. R. Luzung, K. N. Houk, F. D. Toste, *J. Am. Chem. Soc.* **2008**, *130*, 4517–4526.

[12] Selected examples of other transition metal-catalyzed 1,n,m-ene-allene-yne cycloisomerizations: a) J. Liu, A. Ricke, B. Yang, J.-E. Bäckvall, *Angew. Chem. Int. Ed.* **2018**, *57*, 16842–16846; *Angew. Chem.* **2018**, *130*, 17084; b) Y. Kawaguchi, A. Nagata, K. Kurokawa, H. Yokosawa, C. Mukai, *Chem. Eur. J.* **2018**, *24*, 6538–6542; c) D. Cassú, T. Parella, M. Solà, A. Pla-Quintana, A. Roglans, *Chem. Eur. J.* **2017**, *23*, 14889–14899; d) Y. Ohta, S. Yasuda, Y. Yokogawa, K. Kurokawa, C. Mukai, *Angew. Chem. Int. Ed.* **2015**, *54*, 1240–1244; *Angew. Chem.* **2015**, *127*, 1256; e) N. Saito, T. Ichimaru, Y. Sato, *Chem. Asian J.* **2012**, *7*, 1521–1523.

[13] For gold-catalyzed cycloisomerization of 1,n,m-ene-allene-ynes, see: a) G.-Y. Lin, C.-Y. Yang, R.-S. Liu, *J. Org. Chem.* **2007**, *72*, 6753–6757; b) X. Chen, Y. Zhou, J. Jin, K. Farshadfar, A. Ariafard, W. Rao, P. W. H. Chan, *Adv. Synth. Catal.* **2020**, *362*, 1084–1095.

[14] a) C. Chen, Y. Zou, X. Chen, X. Zhang, W. Rao, P. W. H. Chan, *Org. Lett.* **2016**, *18*, 4730–4733; b) W. Rao, J. W. Boyle, P. W. H. Chan, *Chem. Eur. J.* **2016**, *22*, 6532–6536; c) W. Rao, D. Susanti, B. J. Ayers, P. W. H. Chan, *J. Am. Chem. Soc.* **2015**, *137*, 6350–6355.

[15] a) J. H. Lee, F. D. Toste, *Angew. Chem. Int. Ed.* **2007**, *46*, 912–914; *Angew. Chem.* **2007**, *119*, 930; b) H. Funami, H. Kusama, N. Iwasawa, *Angew. Chem. Int. Ed.* **2007**, *46*, 909–911; *Angew. Chem.* **2007**, *119*, 927.

[16] C. Nieto-Oberhuber, S. López, A. M. Echavarren, *J. Am. Chem. Soc.* **2005**, *127*, 6178–6179.

[17] Selected examples of gold-catalyzed [4+2] cycloadditions, see: ref.[13b] and a) X. Chen, J. T. Merrett, P. W. H. Chan, *Org. Lett.* **2018**, *20*, 1542–1545; b) L. Zhou, M. Zhang, W. Li, J. Zhang, *Angew. Chem. Int. Ed.* **2014**, *53*, 6542–6545; *Angew. Chem.* **2014**, *126*, 6660; c) Z.-Y. Han, D.-F. Chen, Y.-Y. Wang, R. Guo, P.-S. Wang, C. Wang, L.-Z. Gong, *J. Am. Chem. Soc.* **2012**, *134*, 6532–6535; d) H. Kusama, Y. Karibe, Y. Onizawa, N. Iwasawa, *Angew. Chem. Int. Ed.* **2010**, *49*, 4269–4272; *Angew. Chem.* **2010**, *122*, 4365; e) C. Nieto-Oberhuber, P. Pérez-Galán, E. Herrero-Gómez, T. Lauterbach, C. Rodríguez, S. López, C. Bour, A. Rosellón, D. J. Córdenas, A. M. Echavarren, *J. Am. Chem. Soc.* **2008**, *130*, 269–279.

[18] Deposition Number 1981291 (**2o**) contains the supplementary crystallographic data for this paper. These data are provided free of charge by the joint Cambridge Crystallographic Data Centre and Fachinformationszentrum Karlsruhe Access Structures service www.ccdc.cam.ac.uk/structures.

[19] B. Bolte, Y. Odabachian, F. Gagosz, *J. Am. Chem. Soc.* **2010**, *132*, 7294–7296.

Received: June 17, 2020

© 2020 Wiley-VCH GmbH